

ASSIGNMENT #1

Course: CHE 301. CHEMICAL ENGINEERING THERMODYNAMICS
Term: Spring 2020
Professor: Dr. Michael Caracotsios
Office: EIB 244
E-mail: mcaracot@uic.edu

Your name: Joseph Bilderback
Due Date: Wednesday January 22 by 5:00pm in my office

Number of points.....100
Your score..... 92

Note: All problems are worth 100 points. You final score will be the average value

95/100

Problem 1A

To prove that $\left(\frac{\partial C_v}{\partial v}\right)_T = T \left(\frac{\partial^2 P}{\partial T^2}\right)_v$ we start with

$$U = U(T, V)$$

Taking the total derivative of $U(T, V)$ gives

$$(1) \quad dU = \left(\frac{\partial U}{\partial T}\right)_V dT + \left(\frac{\partial U}{\partial V}\right)_T dV$$

We know that (2) $\left(\frac{\partial U}{\partial T}\right)_V = C_v$

From the first law of thermodynamics, we know the fundamental relation:

$$(3) \quad \Delta U = \delta Q - \delta W$$

The differential form of equation (3) is as follows:

$$(4) \quad dU = T dS - P dV$$

Taking the partial derivative of equation (4) with respect to V while holding T constant gives

$$(5) \quad \left(\frac{\partial U}{\partial V}\right)_T = \left(\frac{T \partial S - P \partial V}{\partial V}\right)_T = T \left(\frac{\partial S}{\partial V}\right)_T - P \quad \text{OK!}$$

proof? (-5)

Substituting the Maxwell Relation $\left(\frac{\partial S}{\partial V}\right)_T = \left(\frac{\partial P}{\partial T}\right)_V$ into equation (5) gives

$$(6) \quad \left(\frac{\partial U}{\partial V}\right)_T = T \left(\frac{\partial P}{\partial T}\right)_V - P$$

Substituting equations (2) and (6) into equation (1) gives us the expression

$$(7) \quad dU = C_v dT + \left(T \left(\frac{\partial P}{\partial T}\right)_V - P\right) dV$$

Recalling the rule that of a given equation of the form

$$Mdx + Ndy$$

The partial derivatives of them will be equivalent

$$\left(\frac{\partial M}{\partial y}\right)_x = \left(\frac{\partial N}{\partial x}\right)_y$$

Thus, taking the partial derivative with respect to V while holding T constant of equation (7) gives

$$\left(\frac{\partial Cv}{\partial V}\right)_T = T \left(\frac{\partial^2 P}{\partial^2 T}\right)_V + \left(\frac{\partial P}{\partial T}\right)_V - \left(\frac{\partial P}{\partial T}\right)_V$$

The two $\left(\frac{\partial P}{\partial T}\right)_V$ terms cancel, and we are left with

$$\left(\frac{\partial Cv}{\partial V}\right)_T = T \left(\frac{\partial^2 P}{\partial^2 T}\right)_V$$

Thus proving the identity.

Problem 1B

<u>Cp</u>	<u>Heat Capacity under constant pressure, cal/(mol K)</u>
<u>T</u>	<u>Temperature, K</u>
<u>A</u>	<u>7.953090666</u>
<u>B</u>	<u>19.09166906</u>
<u>C</u>	<u>2086.9</u>
<u>D</u>	<u>9.936466991</u>
<u>E</u>	<u>991.96</u>
<u>P1</u>	<u>1 bar</u>
<u>P2</u>	<u>30 bar</u>

In order to find the final temperature of the gas using the secant method, we have to be able to integrate

$$(1) \quad dS = Cp \frac{dT}{T} - R \frac{dP}{P}$$

$$(2) \quad Cp = A + \left(\frac{B \left(\frac{C}{T} \right)^2}{\sinh \left(\frac{C}{T} \right)^2} + D \frac{\left(\frac{E}{T} \right)^2}{\left(\cosh \left(\frac{C}{T} \right) \right)^2} \right) \frac{dT}{T}$$

Substitute equation (2) into equation (1)

$$dS = \left(A + \frac{B \left(\frac{C}{T} \right)^2}{\sinh \left(\frac{C}{T} \right)^2} + D \frac{\left(\frac{E}{T} \right)^2}{\left(\cosh \left(\frac{C}{T} \right) \right)^2} \right) \frac{dT}{T} - R \frac{dP}{P}$$

Since the process is isentropic, the change in entropy is zero

$$0 = \left(A + \frac{B \left(\frac{C}{T} \right)^2}{\sinh \left(\frac{C}{T} \right)^2} + D \frac{\left(\frac{E}{T} \right)^2}{\left(\cosh \left(\frac{C}{T} \right) \right)^2} \right) \frac{dT}{T} - R \frac{dP}{P}$$

Separating the T and P variables gives the expression

$$R \frac{dP}{P} = A \frac{dT}{T} + B \left(\frac{\left(\frac{C}{T} \right)^2}{\sinh \left(\frac{C}{T} \right)^2} \right) \frac{dT}{T} + D \left(\frac{\left(\frac{E}{T} \right)^2}{\left(\cosh \left(\frac{C}{T} \right) \right)^2} \right) \frac{dT}{T}$$

We can now take the integral of both sides of the equation

$$R \int_{P_1}^{P_2} \frac{dP}{P} = A \int_{T_1}^{T_2} \frac{dT}{T} + B \int_{T_1}^{T_2} \left(\frac{\left(\frac{C}{T} \right)^2}{\sinh \left(\frac{C}{T} \right)^2} \right) \frac{dT}{T} + D \int_{T_1}^{T_2} \left(\frac{\left(\frac{E}{T} \right)^2}{\left(\cosh \left(\frac{C}{T} \right) \right)^2} \right) \frac{dT}{T}$$

value of R? (-2)

The equation must be equal to zero to use the secant method, so we subtract

$-R \left(\ln \left(\frac{P_2}{P_1} \right) \right)$ from both sides of the equation

$$0 = -R \left(\ln \left(\frac{P_2}{P_1} \right) \right) + A \ln \left(\frac{T_2}{T_1} \right) + B \left[\left(\frac{C \coth \left(\frac{C}{T_2} \right)}{T_2} \right) - \ln \left(\sinh \left(\frac{C}{T_2} \right) \right) \right] + D \left[\left(\frac{\left(\frac{E}{T_2} \right)^2}{\left(\cosh \left(\frac{C}{T_2} \right) \right)^2} \right) + \ln \left(\cosh \left(\frac{E}{T_2} \right) \right) \right] - \left\{ B \left[\left(\frac{C \coth \left(\frac{C}{T_1} \right)}{T_1} \right) - \ln \left(\sinh \left(\frac{C}{T_1} \right) \right) \right] + D \left[\left(\frac{\left(\frac{E}{T_1} \right)^2}{\left(\cosh \left(\frac{C}{T_1} \right) \right)^2} \right) + \ln \left(\cosh \left(\frac{E}{T_1} \right) \right) \right] \right\}$$

should define these variables

T2(0)	T2(1)	F(T2(0))	F(T2(1))	T3 ?
173.15	153.15	3.072303	2.095158	110.2668
153.15	110.2668	2.095158	-0.51788	118.7658
110.2668	118.7658	-0.51788	0.072653	117.7202
118.7658	117.7202	0.072653	0.002321	117.6857
117.7202	117.6857	0.002321	-1.1E-05	117.6858
117.6857	117.6858	-1.1E-05	1.55E-09	117.6858
117.6858	117.6858	1.55E-09	-7.5E-14	117.6858
117.6858	117.6858	-7.5E-14	0	117.6858
117.6858	117.6858	0	0	#DIV/0!
117.6858	#DIV/0!	0	#DIV/0!	#DIV/0!
#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!

Utilizing the secant method in

Excel makes the answer converge to 117.6858 K. This answer would be -155.4642 °C

Thus the final temperature of the gas is approximately:

-155.4642 °C

how? (-10)

97/100

Problem 1C

$$\text{Sackur-Tetrode: } S = Nkb \left\{ \ln \left[\frac{V}{N} \left(\frac{4\pi M}{3n^2} \right) \left(\frac{U}{N} \right)^{\frac{3}{2}} \right] + \frac{5}{2} \right\}$$

$$\text{Ideal Gas Law: } PV = RT$$

To derive the ideal gas law from the Sackur-Tetrode equation for the entropy of a monatomic ideal gas, we start with the total differential of $U(S,V)$ and a fundamental relation.

$$(1) \quad dU = \left(\frac{\partial U}{\partial S} \right)_V dS + \left(\frac{\partial U}{\partial V} \right)_S dV$$

$$(2) \quad dU = T dS - P dV$$

Looking at (1) and (2) we can see that

$$(3), (4) \quad \left(\frac{\partial U}{\partial S} \right)_V = T, \quad \left(\frac{\partial U}{\partial V} \right)_S = -P$$

To Simplify the Sackur-Tetrode equation, we can use the fact that the number of Moles times the Boltzmann constant equals the gas constant (R)

$$R = Nkb$$

$$(5) \quad \frac{S}{R} = \ln \left[\frac{V}{N} \left(\frac{4\pi M}{3n^2} \right)^{\frac{3}{2}} \left(\frac{U}{N} \right)^{\frac{3}{2}} \right] + \frac{5}{2}$$

Now if we take the partial of the rearranged Sackur-Tetrode (5) equation with respect to S while holding V constant, it will give us

$$\frac{\partial}{\partial S} \left(\frac{S}{R} \right) = \frac{\partial}{\partial S} \left(\ln(v) - \ln(N) + \ln \left(\frac{4\pi M}{3n^2} \right)^{\frac{3}{2}} + \ln(U)^{\frac{3}{2}} + \frac{5}{2} \right)$$

(capital letter $\neq$ small letter)
 (-1)

$$(6) \quad \frac{1}{R} = \frac{3}{2} \left(\frac{1}{U} \right) \left(\frac{\partial U}{\partial S} \right)_V$$

Substituting relation (3) into equation (6) gives

$$(7) \quad \frac{1}{R} = \frac{3}{2} \left(\frac{1}{U} \right) T$$

Now taking the partial of **(5)** with respect to V while holding S constant will yield the expression

$$\frac{\partial}{\partial V} \left(\frac{S}{R} \right) = \frac{\partial}{\partial V} \left(\ln(v) - \ln(N) + \ln \left(\frac{4\pi M}{3n^2} \right)^{\frac{3}{2}} + \ln(U)^{\frac{3}{2}} + \frac{5}{2} \right)$$

$$(-1) \quad (8) \quad 0 = \frac{1}{V} + \frac{3}{2} \left(\frac{1}{U} \right) \left(\frac{\partial U}{\partial V} \right)_S$$

Substituting relation **(4)** into equation **(8)** gives

$$(9) \quad 0 = \frac{1}{V} - \frac{3}{2} \left(\frac{1}{U} \right) P$$

Solving for U in equations **(7)** and **(9)** gives the expressions

$$U = \frac{3}{2} TR = \frac{3}{2} PV$$

Canceling the 3/2 on both sides gives the ideal gas law

$$PV = RT$$